Security and Privacy for Machine Learning (Honors)

Internal Assessment 1

Problem Formulation and Preliminary Implementation

Which Privacy Layer Stops Which Attack?

An Attack-Grounded Comparison of Central DP, Local DP,
Anonymisation and Compressive Privacy for Census-Income Prediction

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Submitted as an individual integrated research work spanning all five course modules.

Contents

1 Problem Definition and Module Mapping **1**
- 1.1 Problem statement . . . 1
- 1.2 Significance, scope and objectives . . . 2
- 1.3 Mapping to all five modules . . . 2

2 Conceptual Understanding **3**
- 2.1 Core concepts and why they fit . . . 3
- 2.2 What will be implemented, demonstrated, or analysed . . . 3
- 2.3 Methodology / workflow . . . 3

3 Dataset Identification and Preliminary Preparation **3**

4 Initial Implementation and Demonstration **4**
- 4.1 Utility baselines and the deployment-layer attack . . . 4
- 4.2 The "just drop the sensitive column" fallacy . . . 4
- 4.3 Publishing-layer linkage risk . . . 5
- 4.4 Module 2 — DP Naive Bayes . . . 5
- 4.5 Module 3 — LDP frequency oracles . . . 5
- 4.6 Module 4 — Mondrian k-anonymity . . . 5
- 4.7 Module 5 — compressive privacy . . . 6

5 Preliminary Evaluation and Documentation **6**
- 5.1 Observations . . . 6
- 5.2 Evaluation measures (for IA-2) . . . 6
- 5.3 Known limitations and remaining work . . . 6
- 5.4 Individual-contribution statement . . . 7

Rubric-compliance map. This report is organised to mirror the five IA-1 criteria; the table below points each required evidence item to where it appears.

IA-1 criterion	Evidence in this report	Section
Problem & module mapping	Problem statement, objectives, scope, five-module mapping table	§1
Conceptual understanding	Concept note, workflow diagram, implement/demonstrate/analyse split	§2
Dataset preparation	Provenance table, cleaning log, hashes	§3
Initial implementation	Baselines, attack suite, one mechanism per module, code	§4
Preliminary evaluation	Observations, limitations, evaluation plan, contribution statement	§5

1 Problem Definition and Module Mapping

1.1 Problem statement

An organisation trains a binary classifier that predicts whether an individual's income exceeds \$50K from personal census records. The same record is exposed at *four distinct points* in the

machine-learning pipeline, and each exposure invites a different privacy attack with a different natural defence:

1. **Collection** from individuals by an untrusted curator ⇒ attribute disclosure.
2. **Publishing** a record-level table to analysts ⇒ linkage re-identification and homogeneity attacks.
3. **Sharing** features with a third-party ML service ⇒ sensitive-attribute inference.
4. **Deploying** the trained model as a query API ⇒ membership inference.

The course offers a different privacy technology for each layer: local differential privacy (Module 3), k-anonymity and its refinements (Module 4), compressive privacy (Module 5), and central differential privacy (Module 2); Module 1 supplies the threat models used to *measure* them.

1.2 Significance, scope and objectives

In the privacy-preserving ML literature each of these defences is almost always evaluated in isolation, on a different dataset and against a different yardstick, which makes it impossible to answer the practitioner's real question: *given a specific exposure and a specific attack, which mechanism should I deploy, and what does it cost?* The contribution of this work is to place all four defence families on **one dataset, one utility metric and one attack suite**, and to summarise the outcome as a *defence-coverage matrix*.

Objectives. (O1) Build an income classifier and a reusable attack suite (membership inference, attribute inference, linkage). (O2) Implement one representative mechanism from every module. (O3) Measure each mechanism's privacy gain and utility cost on common axes. (O4) In IA-2, assemble the coverage matrix and test a layered combination.

Scope. The study is empirical and single-dataset (UCI Adult, with an IA-2 replication on ACSIncome). Cryptographic approaches (homomorphic encryption, secure multiparty computation, federated learning) are treated analytically, not implemented, because they protect *computation* rather than alter the privacy–utility trade-off this study measures.

Expected outcome. A reproducible benchmark and a coverage matrix stating, per attack, which mechanism neutralises it and the accuracy sacrificed.

1.3 Mapping to all five modules

Table 1 maps every module (and the sub-units that are exercised) to its role in the project. The treatment code is **I**=implemented, **E**=experimentally evaluated, **D**=demonstrated, **A**=analytically discussed.

Table 1: Mapping of the problem to the five syllabus modules.

Module	Concepts used	Role in this project	Code
M1 Privacy considerations	Threats/attacks on ML (1.2); privacy complications (1.1); PPML/PPDM taxonomy (1.3–1.4)	Membership/attribute/linkage attacks that score every defence; "dropping sex is not enough" demo	I/E/D/A
M2 Differential privacy	Mechanisms & properties (2.1–2.2); DP supervised learning (2.3); DP unsupervised (2.4)	Laplace/Gaussian; DP Naive Bayes, DP logistic regression, DP-SGD; DP k-means	I/E
M3 Local DP	LDP mechanisms (3.1–3.2); piecewise (3.3); LDP Naive Bayes (3.4)	GRR/SUE/OUE/OLH, Duchi, Piecewise; LDP-NB classifier at the collection layer	I/E

Module	Concepts used	Role in this project	Code
M4 Privacy-preserving data mining	Protect by modifying input (4.2); publishing (4.3); beyond k-anonymity (4.4); modify output (4.5)	Mondrian k-anonymity, ℓ-diversity, t-closeness; output modification vs MIA	I/E/D
M5 Compressive privacy	Mechanisms of compressive privacy (5.1–5.2); use for ML (5.3); utility accuracy (5.4)	PCA/DCA/privacy-aware DCA before sharing features; utility vs dimension	I/E

2 Conceptual Understanding

2.1 Core concepts and why they fit

Differential privacy (DP). A randomised algorithm M is (ε, δ)-DP if for neighbouring datasets D, D' and any output set S, $\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta$ (Dwork et al., 2006). It bounds what the *released model* reveals about any one record, which is exactly the membership-inference threat at the deployment layer. We use the Laplace/Gaussian mechanisms, objective/output perturbation (Chaudhuri et al., 2011) and DP-SGD (Abadi et al., 2016).
Local DP (LDP). Here each user randomises their value *before* it reaches the curator, so no trusted aggregator is assumed – the right model for the collection layer. Frequency oracles (GRR, OUE, OLH) (Wang et al., 2017) and numeric mean oracles (Duchi (Duchi et al., 2013), Piecewise (Wang et al., 2019)) trade accuracy for a local ε.
k-anonymity family. When a record-level table is *published*, generalisation makes every record indistinguishable from $k-1$ others on its quasi-identifiers (Sweeney, 2002; LeFevre et al., 2006); ℓ-diversity (Machanavajjhala et al., 2007) and t-closeness (Li et al., 2007) repair its homogeneity weaknesses.
Compressive privacy. A linear projection compresses features so that the *utility* label survives but a *sensitive* label and raw reconstruction do not (Kung, 2017) – the defence for the feature-sharing layer.

2.2 What will be implemented, demonstrated, or analysed

Mechanisms that drive the coverage matrix (DP-NB/LR/SGD, DP k-means, the LDP oracles, Mondrian and its refinements, PCA/DCA) are *implemented and experimentally evaluated*. The "remove the sensitive column" fallacy is a *demonstration*. Cryptographic privacy (HE/SMPC/FL) is treated *analytically* because it does not change the accuracy–privacy trade-off that this benchmark isolates.

2.3 Methodology / workflow

Figure 1 shows the common pipeline: the same cleaned dataset feeds every exposure layer; a mechanism is inserted at the matching layer; the attack suite then scores both utility and privacy on identical splits.

3 Dataset Identification and Preliminary Preparation

Source and licence. UCI *Adult* / Census Income (Becker and Kohavi, 1996) (Becker & Kohavi, 1996; DOI 10.24432/C5XW20), released under CC BY 4.0 – lawfully usable with attribution.

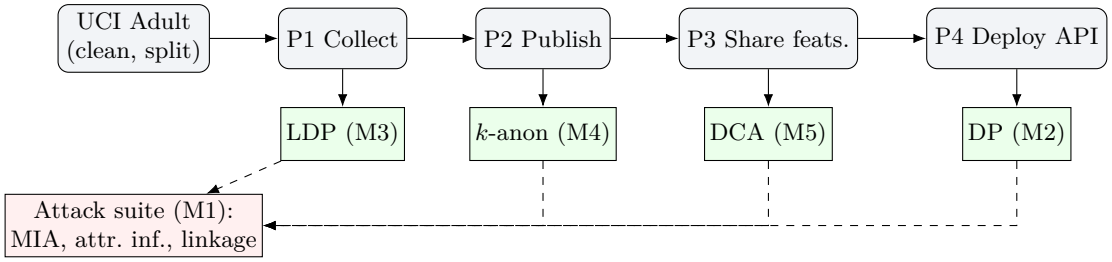


Figure 1: Common evaluation pipeline. Each exposure point P_i receives the module mechanism matched to it; the Module-1 attack suite scores utility and privacy on identical data splits.

Suitability. It is the canonical privacy-ML benchmark: it carries real quasi-identifiers (age, sex, race, marital status, education, native country) and a natural sensitive attribute (sex), so every module's mechanism has a genuine target. Its known limitation – it is a stale 1994 extract with demographic skew (Ding et al., 2021) – motivates the IA-2 replication on ACSIncome.
Preparation log. Whitespace stripped; `?` mapped to missing; rows with any missing value dropped; income mapped to $\{0, 1\}$; the survey weight `fnlwgt` excluded; categoricals one-hot encoded and numerics standardised on *training* statistics only. This yields 31,656 training and 13,566 test records (103 encoded features; positive rate 24.8%), from 48,842 raw rows. Cleaned splits and SHA-256 hashes are recorded in `data/PROVENANCE.md` and `data/processed/processed_manifest.json`.

4 Initial Implementation and Demonstration

All code is in `src/privpipe/` (mechanisms written from scratch; Opacus used only for the DP-SGD accountant) and is exercised by `IA1/scripts_ia1.py`; mechanism correctness is checked in `tests/` (13 passing tests).

4.1 Utility baselines and the deployment-layer attack

Table 2 reports four classifiers and a loss-threshold membership inference attack (MIA) on each. Logistic Regression is the strongest classifier (0.848 accuracy). The Random Forest, which overfits, is the most leak-prone: its MIA AUC is 0.592 (advantage 0.108), empirically linking privacy risk to overfitting (Yeom et al., 2018).

Table 2: Utility baselines and loss-threshold MIA (mean over 3 seeds).

Model	Acc.	F_1	ROC-AUC	MIA AUC	MIA adv.
Logistic Regression	0.848	0.667	0.907	0.497	-0.010
Gaussian Naive Bayes	0.644	0.563	0.854	0.498	-0.002
Random Forest	0.842	0.666	0.892	0.592	0.108
MLP	0.828	0.634	0.875	0.532	0.017

4.2 The "just drop the sensitive column" fallacy

Removing `sex` does not hide it: an adversary reconstructs it from the remaining attributes with accuracy 0.838 versus a majority baseline of 0.677 (a leakage of 0.161). This is the demonstration that motivates genuine mechanisms rather than naive suppression.

4.3 Publishing-layer linkage risk

Table 3 shows that 13.8% of records are *unique* on their quasi-identifiers (achieved $k = 1$), i.e. directly re-identifiable – the risk that k-anonymity must remove.

Table 3: Quasi-identifier uniqueness / linkage risk.

Age bin	Frac. unique	Frac. ≤ 2	Achieved k	#classes
1	0.138	0.191	1	9880
5	0.073	0.105	1	5441
10	0.055	0.080	1	4196

4.4 Module 2 — DP Naive Bayes

Table 4 and Figure 2 give a first ε sweep for a scratch DP Naive Bayes (Laplace noise on class-conditional counts). Utility is near-flat across $\varepsilon \in [0.5, 5]$, indicating the binned-count mechanism is noise-tolerant on this dataset – a first data point for the trade-off curves.

Table 4: DP Naive Bayes ε sweep.

ε	Acc.	Bal. Acc.	F_1	ROC-AUC
0.5	0.767	0.755	0.609	0.835
1.0	0.756	0.757	0.607	0.855
2.0	0.754	0.758	0.607	0.859
5.0	0.753	0.760	0.608	0.860
∞	0.644	0.738	0.563	0.854

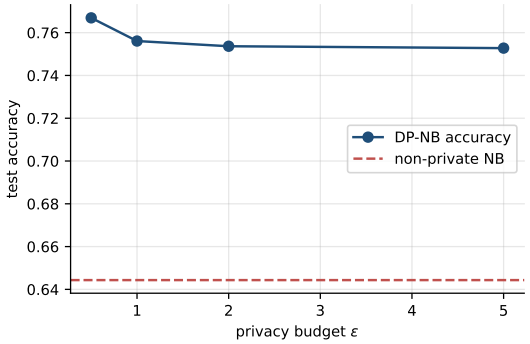


Figure 2: DP-NB accuracy vs ε.

4.5 Module 3 — LDP frequency oracles

Figure 3 and Table 5 estimate the `education` distribution under GRR and OUE. OUE's mean-squared error is one to two orders of magnitude below GRR's at small ε, matching the theory that unary encoding dominates generalised randomised response on larger domains (Wang et al., 2017).

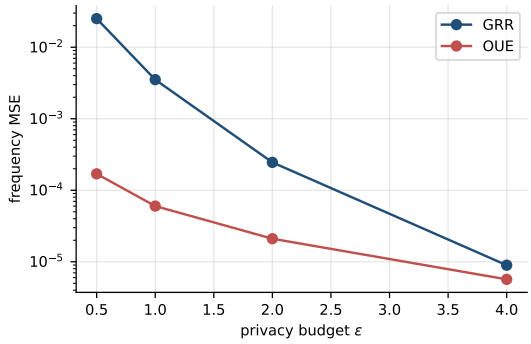


Figure 3: LDP frequency MSE vs ε (log scale).

Table 5: LDP frequency-estimation MSE on `education`.

ε	GRR MSE	OUE MSE
0.5	2.51e-02	1.70e-04
1.0	3.52e-03	6.01e-05
2.0	2.45e-04	2.10e-05
4.0	8.96e-06	5.68e-06

4.6 Module 4 — Mondrian k-anonymity

A scratch Mondrian anonymiser reaches the target $k = 5$ exactly (Table 6), but exposes the classic weakness: five equivalence classes are *homogeneous* in the sensitive attribute (min $\ell = 1$)

and the worst-class t-closeness is high. This is precisely the gap that ℓ-diversity and t-closeness close in IA-2.

Table 6: Preliminary Mondrian k-anonymity ($k = 5$) and its homogeneity weakness.

Property	Value
Target k	5
Achieved k (min class size)	5
#equivalence classes	1123
Min ℓ (distinct-ℓ diversity)	1
Homogeneous classes	5
Records disclosed by homogeneity	0.001
Worst-class t (total-variation)	0.889

4.7 Module 5 — compressive privacy

Table 7 compresses the 103-dimensional features to $p = 10$ and reports both utility (income accuracy) and leakage (how well the sensitive `sex` label survives). PCA preserves *everything*, so sex is recovered at accuracy 1.000; privacy-aware DCA keeps income accuracy high (0.840) while cutting sex recovery to 0.689 – the compressive-privacy trade-off in a single row.

Table 7: Preliminary PCA vs DCA ($p = 10$): utility is income accuracy, leakage is sex accuracy.

Projection (p=10)	Income acc. (utility)	Sex acc. (leakage)	Recon. err.
PCA	0.839	1.000	0.492
DCA	0.844	0.831	0.751
pa-DCA	0.840	0.689	0.780
random	0.790	0.913	0.688

5 Preliminary Evaluation and Documentation

5.1 Observations

- Overfitting drives leakage: the RF's high MIA AUC (0.592) confirms the overfitting–privacy link and justifies DP at the deployment layer.
- Suppression fails (attribute leakage 0.161 above baseline); real mechanisms are needed.
- Each module's preliminary mechanism already shows its signature trade-off: OUE $\gg$ GRR in accuracy; DP-NB noise-tolerant; Mondrian attains k but not ℓ; privacy-aware DCA suppresses the sensitive label at little utility cost.

5.2 Evaluation measures (for IA-2)

Utility: accuracy, balanced accuracy, F_1, ROC-AUC. Privacy: ε (at $\delta = 10^{-5}$), $k/\ell/t$; MIA AUC, advantage and TPR@1%FPR (Carlini et al., 2022); attribute-inference accuracy over baseline; re-identification rate; LDP estimator MSE vs theory. All IA-2 results will be averaged over 5 seeds with mean $\pm$ std.

5.3 Known limitations and remaining work

Single dataset and a US-centric 1994 vintage (addressed by the ACSIncome replication); preliminary mechanisms use fixed hyper-parameters (IA-2 adds full ε sweeps and shadow-model

MIA); the coverage matrix and the layered DCA + DP-SGD combination are the main IA-2 deliverables.

5.4 Individual-contribution statement

I, **Deon Ajit Menezes** (16014223030, B1), independently designed the four-exposure framing and the single-benchmark methodology, and wrote the `privpipe` package from scratch – the data pipeline, the membership / attribute / linkage attacks, the DP Naive Bayes, the LDP frequency and numeric oracles, the Mondrian anonymiser with the ℓ-diversity and t-closeness checks, and the PCA/DCA compressive-privacy module – together with the unit tests and the experiment driver that produced every number above. Opacus is used only as the DP-SGD privacy accountant in IA-2; all other mechanisms are my own implementation.

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